

A Computationally Feasible Framework for Causal Probabilistic Explanation

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Abstract

Causal attribution and explanation are mainstays of philosophical, scientific, and policy analysis. Some current frameworks, however, such as actual causality, suffer from an inability to scale causal reasoning to large problems, while attribution methods that do scale—such as SHAP or LIME—are not causally aware. To address this two-fold problem, we present Probabilistic Causal Impact (PCI). Inspired by Halpern and Pearl’s actual causality and Pearl’s probability of necessary and sufficient causation, PCI provides tractable causal explanations in causal probabilistic machine learning models. By the same token, PCI provides explanations for events in the world, insofar as the model adequately represents real mechanisms. By re-conceptualizing explanation as an estimation problem on an expanded, causal probabilistic model, PCI allows for the use of well-established approximation algorithms, unlocking computationally tractable, truly causal explanations for large models. We test PCI on scaled-up, but canonical test cases for actual causation involving over-determination and undercutting, finding that we recover results deemed intuitive by the literature, and illustrate that the correct behavior is preserved in the continuous version of the problem. We apply PCI to a real-life causal spatio-temporal model with complex machine-learned components trained on millions of data points. In this setting, we compare PCI to the non-causal (yet scalable) attribution method SHAP, and find that our approach yields explanations that agree with domain expert opinion, while SHAP does not.

Keywords: Probabilistic Causal Models, Explanation, Responsibility, Actual Causality, Probability of Necessity and Sufficiency

1. Introduction

Causal attribution is a central reasoning task in many important problems, such as gauging the efficacy of medical treatments (Shepherd et al., 2024), pinpointing drivers of economic growth (Cheshire and Magrini, 2009), root cause analysis in automated systems (Papageorgiou et al., 2022), or assessing the impact of public policy (Heckman and Pinto, 2022). In general, we may want to understand why an adverse outcome occurred to prevent it from happening again, or understand how and why a predicted, counterfactual positive outcome can be brought about. However, in the absence of scalable, formal causal attribution tools, analysts often resort to either ad-hoc methods, such as mechanistic reasoning and finding strong correlation, historical comparison, domain expertise, or non-causal attribution like Granger causality (Granger, 1969) and SHAP (Lundberg and Lee, 2017). Even researchers with an advanced command of statistical and experimental methods may not possess a thorough understanding of how to ascribe cause (Shepherd et al., 2024), and would thus benefit from scalable automation. Here we present a tool for accurate and tractable causal attribution that can be applied to a wide array of causal, probabilistic, machine-learned models.

Current methods for causal attribution that focus on token-level explanations of particular events, such as actual causality (Halpern, 2016), involve intractable combinatorics without clear approximating machinery (Eiter and Lukasiewicz, 2002). Explanation methods that *do* scale are either

non-causal (Lundberg and Lee, 2017) or amount to local, gradient-based sensitivity analyses (Butler et al., 2022). For example, Lundberg and Lee’s SHAP method may assign high attribution to a feature simply because it correlates with a true cause, not because altering said feature would change the outcome. Butler et al. in their Differential Causal Effect (DCE) rely on gradients for attribution, which requires model differentiability, and does not capture effects that require more than infinitesimal changes in potential causes (admittedly, the authors are only after rates of change of treatment, which is conceptually different from responsibility attribution).

In contrast, we introduce probabilistic causal impacts (PCI), which exploit probabilistic understandings of context and causation to frame attribution as a turnkey estimation problem. In particular, we generalize Pearl’s formulation of the “probability of necessary and sufficient causation” (Pearl, 2022) for arbitrary variable domains, and additionally incorporate causal context in the form of “witnesses” as understood by actual causality researchers (Halpern, 2016). Inspired by the study of “descriptive normality” (Halpern and Hitchcock, 2015; Icard et al., 2017) — how statistical probabilities of counterfactuals influence causal attributions — we characterize attribution as the expectation of a customizable causal impact function taken with respect to user-defined distributions that assign probabilities to counterfactual worlds. PCI combines a variety of intuitions from the literature, including taking expectations over exogenous inputs (cf. Halpern (2016) on “blame”, def. 6.2.1; and implicitly by Pearl’s (2022) probabilities of causation), weighting by the probability of alternative values for the supposed cause (Beckers and Vennekens, 2015, def. 8), aggregating the probabilities of all valid witness worlds (Beckers and Vennekens, 2015, §6.2), or weighting by the complexity of the least complex witness set (Halpern, 2016, on “responsibility”, def. 6.1.6). In Appendix ??, we discuss related concepts in more detail. Unfortunately, taken individually, none of these existing approaches achieves the scalability and generality required for modern causal attribution needs, thereby necessitating the development of PCI.

The mathematical and computational tooling we develop in this paper is designed for a wide range of causal probabilistic models; we place no strong requirements on the form of the model, distributions, structural equations, or types of variables that can be used. Indeed, such models can comprise standard machine learning components like neural networks or Gaussian processes. With this breadth of application, and the turnkey nature of our measure of causal attribution, PCI hopes to provide *truly causal* attribution and explanation tooling for a wide array of users and applications.

To validate and better understand the approach, and to illustrate how our method scales to far larger models than does exact actual causality computation, we take the following steps:

revise this list once paper done

1. We draw theoretical connections to existing (but intractable) attribution strategy, actual causality in Section ??, where we prove that, under certain constraints, the diagnoses offered by actual causality and by PCI coincide.
2. In Section ??, we test PCI on scaled-up, but canonical failure cases for actual causation involving overdetermination and undercutting, finding that we recover results deemed intuitive by the literature,
3. In Section ?? we point also to the correct behavior in the continuous version of the problem.
4. We also delegate a more extensive conceptual discussion of the relation between PCI, actual causality, and probabilities of necessity, sufficiency and necessity and sufficiency to Appendix ??.

5. In Section ?? we describe results on a causal spatio-temporal model with complex machine-learned components trained on millions of data points. In this setting, we compare to the non-causal but scalable attribution method SHAP, and find that our approach yields explanations which agree with domain expert opinion, while SHAP does not.
6. Additionally, in Appendix ??, we compare the local, sensitivity-analysis-style attribution from [Butler et al. \(2022\)](#), demonstrating cases where PCI better reveals causality.

2. Causal Impact: motivations

The central difficulty in causal attribution is not computing counterfactuals — it is knowing which counterfactuals to compute, and recognising when a cause is blocked by context rather than absent altogether. The closest predecessors of this approach, taken individually, fail to capture this: actual causality handles context sensitivity but is computationally intractable and not general enough; the probability of necessity and sufficiency scales well, but lacks context sensitivity. This section builds intuition for both limitations through a running example, setting up the design choices that motivate PCI in Section 3.

2.1. Actual Causality

Consider the following (deterministic, discrete) model:

Example 1 (Bob at Old Boys’ Club Bank) *Bob applied for a home mortgage loan with the local branch of Old Boys’ Club Bank (OBCB), and his application was denied. The OBCB policy requires applicants to pass a credit score check, but it only performs this check if the applicant is male; all non-male applicants are rejected outright. Thus, the procedure is:*

$$\begin{aligned} \text{check} &= (\text{gender} = \text{male}) \\ \text{check-failed} &= \text{check} \wedge (\text{credit} = \text{bad}) \\ \text{loan} &= \neg \text{check-failed} \wedge (\text{gender} = \text{male}) \end{aligned}$$

Why was Bob’s loan application denied? An intuitive explanation is that his credit was bad—had it been good, the outcome would have been different. In contrast, his gender does not appear to be responsible for the rejection: had it been different, his application still would have been denied (albeit this time due to $\text{check} = F$).

This suggests a counterfactual approach to feature responsibility attribution: perhaps we can identify the features responsible for a particular outcome by asking counterfactual questions—would the outcome have been different if a specific feature had a different value? This is called the *but-for clause*: Bob wouldn’t have been denied the loan but for his bad credit.

Concretely, consider a model M with *exogenous variables* $\mathbf{U}$, which are the source of stochasticity, and *endogenous variables* $\mathbf{V}$, which are influenced by other variables in the model. Let $\mathbf{S} \subseteq \mathbf{V}$ be the variables we suspect to be causally responsible (call them *causal suspects*), and $Y \in \mathbf{V}$ be an outcome of interest. Suppose we consider only one suspect $S \in \mathbf{S}$ and mark factual values with $*$. Under a setting $\mathbf{U} = \mathbf{u}$ of the exogenous variables, M produces $S = s^*$ and $Y = y^*$, denoted $(M, \mathbf{u}) \models S = s^*, Y = y^*$. We call this the *factual scenario*. An intervened model where a variable S is set to an alternative value s' is denoted $(M, \mathbf{u})[S \leftarrow s']$.

A first attempt at defining a counterfactual notion of causal responsibility is:

Definition 1 (But-for, single) $S = s^*$ is causally responsible for $Y = y^*$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models S = s^*, Y = y^*$ and $(M, \mathbf{u})[S \leftarrow s'] \not\models Y = y^*$ for some $s' \neq s^*$.

In a given context, S is causally responsible for an outcome only when intervening on S with some alternative value would change that outcome. Bob's credit score satisfies this condition: had his credit score been higher, the loan might have been approved. However, this cause test is too coarse.

Consider another case:

Example 2 (Alice at OBCB) *Alice is a woman with bad credit. She applies for a loan with OBCB and is rejected.*

Why was Alice denied the loan? Under the but-for clause for single cause candidates, neither her gender nor her credit score are responsible. No matter how high her credit score, her application would still have been denied because the bank does not loan to women. Had her gender been different, she would still have been denied for having bad credit.

To address this example, we could consider sets of factors as causes:

Attempt 1 (But-for, sets) Variables $\mathbf{C} \subseteq \mathbf{S}$ with factual values $\mathbf{C} = \mathbf{c}^*$ are causally responsible for $Y = y^*$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^*, Y = y^*$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}'] \not\models Y = y^*$ for some $\mathbf{c}'$ everywhere different from $\mathbf{c}^*$, where no proper subset of $\mathbf{C}$ has this property.

The minimality condition reflects our preference for explanations that include only relevant properties.

Under this generalization, the set $\mathbf{c}^* = \{\text{gender} = \text{female}, \text{credit} = \text{bad}\}$ is jointly responsible for Alice's application being denied because there exists $\mathbf{c}' = \{\text{gender} = \text{male}, \text{credit} = \text{good}\}$ that would change the decision and $\mathbf{c}^*$ is minimal. But what can we say about the individual features? We could say that the whole set bears responsibility but individual features do not, or that any feature in a cause set is partially and equally accountable. Both approaches have issues, as they imply the features are equally responsible. This conflicts with our intuition: OBCB did not evaluate Alice's credit score, so her denial was caused by her gender.¹

Prior work suggests a way out of this difficulty: hold some of the context in which counterfactuals are evaluated fixed, calling such fixed variables *witnesses*. Let us focus on the mediator *check-failed* between *gender* and *loan*. If we fix this variable at the factual value by intervening, then counterfactually, Alice would have been granted the loan if she were male. However, there is no context fixed at the actual value such that, had her credit score been higher, she would have been granted the loan. This motivates the following formalization:

Definition 2 (Actual cause) Let $\mathbf{C} \subseteq \mathbf{S}$ be a cause set, $\mathbf{W} \subseteq \mathbf{V}$ be a set of possible witnesses. $\mathbf{C} = \mathbf{c}^*$ is causally responsible for $Y = y^*$ in $(M, \mathbf{u})$ just in case there is a $\mathbf{T}$ (an actual witness set to be held fixed) s.t. $\mathbf{T} \subseteq \mathbf{W}, \mathbf{T} \cap \mathbf{C} = \emptyset$ and $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^*, Y = y^*$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^*] \not\models Y = y^*$, for some $\mathbf{c}'$ distinct everywhere from $\mathbf{c}^*$, where no proper subset of $\mathbf{C}$ satisfies this property.

1. Examples of this sort, involving both over-determination and undercutting of causal impact, abound. Suppose Jim is both stabbed and poisoned and dies as a result. Because stabbing works quickly, it kills Jim before the poison takes effect. In this case, it's the stabbing that caused his death. Yet analogous but-for clauses fail to break the symmetry here as well.

This definition extends but-for causality with a witness set $\mathbf{T}$ that is fixed to its factual value by an intervention. Under this definition, to determine the cause of an outcome, we need to find a cause set $\mathbf{C}$, alternative values $\mathbf{c}'$ (trivial if the variables are binary and less so if they are categorical or continuous), and a witness set $\mathbf{T}$. $\mathbf{C}$ and $\mathbf{T}$ are drawn from sets of size at most $2^{|\mathbf{V}|}$, so a brute-force computation of these sets is expensive.

Putting computational difficulties aside, we have so far considered deterministic models, but would like to capture probabilistic models as well, so we now consider a stochastic variant of the OBCB problem.

2.2. Probability of Necessity and Sufficiency

Example 3 (OBCB, stochastic) *The bank performs credit checks for a randomly selected 20% of women and skips checks for a randomly selected 10% of men. Unchecked applicants are rejected. Furthermore, if a man’s credit check fails, he is nevertheless accepted 5% of the time and if a woman’s credit check succeeds, she is denied 10% of the time. We assume that the population is evenly divided into male and female and good and bad credit. Formally, the model, using 1 for male and 1 for good credit, is:*

	<i>check-failed=0</i>	<i>check-failed=1</i>
<i>loan-prob = gender=0</i>	0.9	0
<i>gender=1</i>	1	0.05

$$\begin{aligned}
 & \textit{credit} \sim \text{Bern}(0.5) \\
 & \textit{gender} \sim \text{Bern}(0.5) \\
 & \textit{check} \sim \text{Bern}((1 - \textit{gender}) \cdot 0.2 + \textit{gender} \cdot 0.9) \\
 & \textit{check-failed} = \textit{check} \cdot (1 - \textit{credit}) \\
 & \textit{loan-if-checked} \sim \text{Bern}(\textit{loan-prob}[\textit{gender}, \textit{check-failed}]) \\
 & \textit{loan} = \textit{loan-if-checked} \cdot \textit{check}
 \end{aligned}$$

The witness mechanism works well for deterministic models, but actual causality in its standard form offers no straightforward way to handle probabilistic ones. Before extending the witness idea to the stochastic setting, it is instructive to examine a probabilistic approach that scales more naturally: Pearl’s probabilities of necessity and sufficiency (Pearl, 2022). This will let us identify what probabilistic machinery is needed, and what context-sensitivity is lost when witnesses are absent — motivating the synthesis that PCI provides. The following definitions allow for probabilistic models, but assume that all variables are Boolean (PCI will relax this assumption, allowing for continuous variables). Pearl considers two ways that a cause may relate to an effect: it may be necessary or sufficient for the effect. A cause is necessary if, counterfactually absent the cause, the effect is unlikely to occur; it is sufficient if the effect is counterfactually likely in the presence of the cause. We now formalize these ideas and discuss them in the context of the running example.

Definition 3 (Probability of Necessity) *Let the probability of necessity be the probability that the outcome $Y = y$ would not have occurred in the absence of $C = c^*$, given that both occurred:*

$$PN(C = c^*, Y = y^*) = P(Y_{c'} \neq y^* \mid C = c^*, Y = y^*),$$

where $Y_{c'}$ is the value of Y after intervening on C with $c' \neq c^*$.

The probability of necessity is less specific to a particular situation than the actual cause test discussed above. For example, we can consider the probability that being male causes someone to fail to get a loan, but we can't consider the probability that being male causes Bob specifically (who we know has bad credit) to fail to get a loan.

This is messy, cleanup

To compute

$$PN(\text{gender} = \text{male}, \text{loan} = F) = P(\text{loan}_{\text{gender}=\text{female}} = T \mid \text{gender} = \text{male}, \text{loan} = F),$$

we first determine that $P(\text{credit}^* = \text{good}) = \frac{20}{211} \approx 0.09$ using the condition (men with good credit are rarely rejected). The probability of getting a loan in the counterfactual case is then $P(\text{credit}^* = \text{good}) \times 0.2 \times (1 - 0.1) = \frac{18}{1055} \approx 0.02$. We find that $PN(\text{gender} = \text{female}, \text{loan} = F) \approx 0.43$ and $PN(\text{credit} = \text{bad}, \text{loan} = F) \approx 0.53$; bad credit has a notably higher necessity score than being female. Both exceed $PN(\text{gender} = \text{male}) \approx 0.02$, reflecting that female gender and bad credit are each more likely to necessitate rejection. The value for $\text{gender} = \text{female}$ remains below 0.5 because many rejected females also have bad credit, which would prevent loan approval even in the counterfactual world where gender is male. However, we know that having low credit and being female are actually quite likely to cause a loan rejection on their own. For Alice specifically, $PN(\text{gender} = \text{female}, \text{loan} = F) = P(\text{loan}_{\text{gender}=\text{male}} = T \mid \text{gender} = \text{female}, \text{credit} = \text{bad}) = 0.045$: if she were male, a credit check would be performed (90% probability), and despite bad credit she would have a 5% chance of approval. For her credit, the counterfactual is $P(\text{loan}_{\text{credit}=\text{good}} = T \mid \text{gender} = \text{female}, \text{credit} = \text{bad}) = 0.2 \times 0.9 = 0.18$. Bob's credit necessity score is 0.90: had his credit been good, a check would have been performed and he would almost certainly have been approved.

Notice how Bob's PN score for credit gets high necessity (0.9), while Alice's is lower (0.18). This partially captures the intuition that Bob's bad credit is more responsible for his rejection than Alice's. But while Bob's PN score for gender is 0, which perhaps partially captures the intuition that his gender is not responsible for his rejection, Alice's PN score for gender is only 0.045, which is not that different, and does not capture the intuition that her gender did play a significant role in her rejection. To see the full picture, we must consider another quantity: the probability of sufficiency.

Definition 4 (Probability of Sufficiency) Let the probability of sufficiency be the probability that the outcome $Y = y^*$ would have occurred in the presence of $C = c^*$, given that $C = c^*$ did not occur:

$$PS(C = c^*, Y = y^*) = P(Y_{c^*} = y^* \mid C = c', Y = y')$$

where Y_{c^*} is the value of Y after intervening on C with $c^* \neq c'$.

To query whether the factual cause was sufficient, we condition on the counterfactual: we assume the cause did *not* take its factual value and the outcome did *not* occur, then intervene to restore the factual cause value and ask whether the outcome now occurs. Intuitively, this asks: if the cause had been absent and things had gone differently, would reinstating the cause have brought the outcome about? Let's look at the resulting values.

Computing the probabilities of sufficiency, we find that $PS(\text{gender} = \text{male}, \text{loan} = F) \approx 0.1$, $PS(\text{gender} = \text{female}, \text{loan} = F) \approx 0.83$, and $PS(\text{credit} = \text{bad}, \text{loan} = F) \approx 0.96$. These values

tell us that being female or having bad credit are both on their own sufficient for a loan rejection, with bad credit being slightly more sufficient. For Alice, both variables achieve $PS = 1$: being female is fully sufficient to cause denial as the model offers no path to approval for women with bad credit, and having bad credit forecloses approval entirely, since $P(\text{loan} = 1 \mid \text{gender} = \text{female}, \text{credit} = \text{bad}) = 0$. Bob’s gender yields an undefined PS: computing it requires conditioning on Bob being female, having bad credit, and succeeding, a conjunction the model assigns probability zero, making the conditional undefined. Bob’s credit yields $PS = 0.955$: restoring bad credit leads to denial in 95.5% of worlds, not certainty, because even with bad credit there remains a 4.5% chance of approval (90% probability of being checked, times the 5% exception rate for males who fail the check). These cases illustrate that PS can fall short of 1 even when a factor is strongly sufficient, and can be undefined when the conditioning event is near-impossible.

To somehow combine insights provided by the two scores, we would like to find causes that are both necessary and sufficient. This notion is formalized as follows:

Definition 5 (Probability of necessity and sufficiency) *Let the probability of necessity and sufficiency be the probability that the outcome $Y = y^*$ would have occurred in the presence of $C = c^*$ and that it would not have occurred under the intervention $C = c'$:*

$$PNS(C = c^*, Y = y^*) = P(Y_{c^*} = y^*, Y_{c'} \neq y^*)$$

When we compute these probabilities for our example, we find that $PNS(\text{gender} = \text{male}, \text{loan} = F) \approx 0.01$, $PNS(\text{gender} = \text{female}, \text{loan} = F) \approx 0.39$, and $PNS(\text{credit} = \text{bad}, \text{loan} = F) \approx 0.52$. While these probabilities reflect the relative causal impact of *gender* and *credit*, they do not provide entirely satisfying answers. Table 1 reveals the problem directly: for Alice, $PNS(\text{gender}) = 0.045 < PNS(\text{credit}) = 0.18$, so PNS ranks credit as more causally responsible than gender. Yet the bank never evaluated Alice’s credit—the *check* variable was *F* throughout. PNS has no mechanism to register this: it asks counterfactually what would happen if credit were different, without fixing the mediating variable *check-failed* that actually blocked credit’s causal path.

As an attempt to help Alice better understand her particular situation, we might consider a modification of PNS with additional conditioning:

$$PNS_c(C = c^*, Y = y^* \mid F = f^*) = P(Y_{c^*} = y^*, Y_{c'} \neq y^* \mid F = f^*)$$

Using this definition, we can compute $PNS_c(\text{gender} = \text{female}, \text{loan} = F \mid \text{credit} = \text{bad}) = 0.045$ and $PNS_c(\text{credit} = \text{bad}, \text{loan} = F \mid \text{gender} = \text{female}) = 0.18$. Conditioning does not solve the problem described above—Alice’s credit was in reality never evaluated—because the interventions override the mediating variable that tracks whether her credit has been checked. When we compute $PNS_c(\text{credit} = \text{bad}, \text{loan} = F \mid \text{gender} = \text{female})$, conditioning on gender being female is consistent with the factual—but when we then intervene to set credit to good, this intervention propagates downstream and changes *check-failed*, overriding what we learned from conditioning on Alice’s factual outcome. The context we fixed is undone by the very intervention meant to probe it. We will not consider this use of conditioning further, and this observation motivates the need for intervention-based (as opposed to conditioning-based) context-sensitivity, which is what the witness mechanism provides.

Table 1: Individual-level PN , PS , and PNS for Alice and Bob in Example 3, with $loan = F$ as the outcome. Values are conditioned on each subject’s factual context (female, bad credit for Alice; male, bad credit for Bob). A dash marks an entry that is undefined because the conditioning premises are inconsistent with the model.

	Alice			Bob		
	PN	PS	PNS	PN	PS	PNS
<i>gender</i>	0.045	1.00	0.045	0	—	0
<i>credit</i>	0.18	1.00	0.18	0.90	0.955	0.86

2.3. Path forward

What should we conclude from this investigation of actual causality and the probability of necessity and sufficiency? First, we need a way to treat probabilistic models, and the probability of necessity and sufficiency gives us a tool for doing that. However, PNS is not as specific as we would like because it does not capture relevant features of individual units that are not part of the cause. Actual causality offers more specificity, but requires modification to deal with probability and continuous variables, and is computationally expensive to compute.

PCI addresses both limitations by combining the two approaches. To see the intuition, consider Alice’s case once more. When PCI evaluates *gender*’s causal impact on her denial, it considers counterfactual worlds in which her gender is changed, while drawing witness sets from a distribution that includes *check-failed* at its factual value of F — the credit check was never performed. In those worlds, changing gender to male, with *check-failed* held fixed at F , leads to loan approval: a large counterfactual shift, and so high attribution. When PCI evaluates *credit*’s causal impact, it searches for witness sets that, held fixed, would make a change to good credit matter for Alice — but no such context exists, because the credit check was never run. The distribution over witness sets therefore assigns low weight to any world in which credit is relevant for Alice, and her credit receives low attribution. Unlike actual causality, PCI does not ask whether *some* witness set validates a counterfactual dependency; it integrates over the distribution of all witness sets, weighting each by its probability. Unlike PNS , it inherits the context-sensitivity of actual causality through this integration. The formal definitions in Section 3 make this expectation precise — but the intuition is the one we have just seen: PCI inherits the witness mechanism of actual causality and the probabilistic language of PN/PNS , and combines them into a tractable estimation problem.

3. Causal Impact: definitions

Explanation and causal attribution seek to connect a potentially causal *event* $X = x$ to an outcome event $Y = y$.² In this section, we present a formal development, but the reader is invited to familiarize themselves with Appendix ?? for a pedagogical explanation that connects our approach to Pearl’s probabilities of causation and Halpern’s actual causality.

To ground the development, we return to Example 3 and use Alice and Bob as running illustrations throughout this section. Both are rejected for a loan; the question is which of their

2. This is quite different from type-level causal queries, where one is interested in some group-level attribution, as when one estimates average treatment effect or conditional average treatment effect.

attributes—*gender* and *credit*—was causally responsible for each rejection, and to what degree. While the example is discrete, the same machinery applies to continuous and mixed-type variables; a continuous illustration is given in Section ??.

Add cross-reference to continuous example.

Let $R(V \rightsquigarrow loan \mid \text{person})$ denote the causal responsibility of variable V for the loan outcome at the individual level. The following qualitative desiderata encode what an intuitively correct attribution should look like:

$$\mathbf{D-A1} \quad R(\text{gender} \rightsquigarrow loan \mid \text{Alice}) > 0$$

$$\mathbf{D-A2} \quad R(\text{credit} \rightsquigarrow loan \mid \text{Alice}) > 0$$

$$\mathbf{D-A-rank} \quad R(\text{gender} \rightsquigarrow loan \mid \text{Alice}) > R(\text{credit} \rightsquigarrow loan \mid \text{Alice})$$

$$\mathbf{D-B1} \quad R(\text{gender} \rightsquigarrow loan \mid \text{Bob}) \geq 0$$

$$\mathbf{D-B2} \quad R(\text{credit} \rightsquigarrow loan \mid \text{Bob}) > 0$$

$$\mathbf{D-B-rank} \quad R(\text{credit} \rightsquigarrow loan \mid \text{Bob}) > R(\text{gender} \rightsquigarrow loan \mid \text{Bob})$$

$$\mathbf{D-comp} \quad R(\text{gender} \rightsquigarrow loan \mid \text{Alice}) > R(\text{gender} \rightsquigarrow loan \mid \text{Bob})$$

D-A1 and **D-A2** both require positive attribution for Alice because two distinct causal routes were available: gender filtered her out at the checking stage without her credit ever being evaluated, while bad credit would have mattered had she been checked. Neither is causally inert. **D-A-rank** is the substantive constraint: because Alice never underwent a credit evaluation, gender’s causal role is immediate and structural, whereas credit’s is purely hypothetical. A method that ranks credit above gender for Alice is misattributing the cause of her rejection.

For Bob, **D-B2** reflects that the credit check occurred and directly produced the rejection: the causal chain $\text{credit} \rightarrow \text{check-failed} \rightarrow \text{loan}$ was activated. **D-B1** is deliberately weak—being male gave Bob a 90% probability of being checked, so his gender opened the evaluation that rejected him rather than causing the rejection itself. **D-B-rank** follows: credit is the proximate cause; gender, if anything, slightly facilitated his evaluation.

D-comp captures the cross-individual asymmetry that is hardest for naive methods to express: gender is more causally responsible for Alice’s rejection than for Bob’s, because for Alice it blocked the evaluation mechanism entirely, while for Bob it opened it. We will verify at the end of this section that PCI satisfies all seven desiderata, while PNS fails **D-A-rank**.

We define causes in the context of probabilistic structural causal models (Pearl, 2009).

Definition 6 (Probabilistic Structural Causal Model) A probabilistic structural causal model is a tuple

$$M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle,$$

where $\mathbf{V} = \{V_1, \dots, V_n\}$ is a finite set of endogenous variables, $\mathbf{U} = \{U_1, \dots, U_n\}$ is a set of exogenous variables, and $\mathbf{F} = \{f_1, \dots, f_n\}$ is a set of structural assignments

$$V_i := f_i(\mathbf{Pa}_i, U_i), \quad i = 1, \dots, n,$$

with $\mathbf{Pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$ denoting the parents of V_i . The distribution $P_{\mathbf{U}}$ is a probability measure on $\text{dom}(\mathbf{U})$, and the assignments in $\mathbf{F}$ are assumed to be acyclic and to have a unique solution for $\mathbf{V}$ given $\mathbf{U}$.³

Informally, a PSCM encodes which variables influence which others, through what mechanisms, and subject to what randomness. The structural equations specify each variable as a deterministic function of its causal parents and local noise; the exogenous distribution captures all irreducible uncertainty in the system. Together they make every counterfactual question well-posed and computable.

The OBCB model (Example 3) is a PSCM with endogenous variables $\mathbf{V} = \{\text{gender}, \text{credit}, \text{check}, \text{check-failed}, \text{loan}\}$ and exogenous variables $\mathbf{U} = \{U_{\text{gender}}, U_{\text{credit}}, U_{\text{check}}, U_{\text{loan}}\}$ drawn independently from $\text{Unif}(0, 1)$, and structural assignments

$$\begin{aligned} \text{gender} &:= \mathbb{I}\{U_{\text{gender}} \leq 0.5\}, & \text{credit} &:= \mathbb{I}\{U_{\text{credit}} \leq 0.5\}, \\ \text{check} &:= \mathbb{I}\{U_{\text{check}} \leq (1 - \text{gender}) \cdot 0.2 + \text{gender} \cdot 0.9\}, \\ \text{check-failed} &:= \text{check} \cdot (1 - \text{credit}), \\ \text{loan} &:= \mathbb{I}\{U_{\text{loan}} \leq \text{loan-prob}[\text{gender}, \text{check-failed}]\} \cdot \text{check}. \end{aligned}$$

For Alice ($\text{gender} = 0$, $\text{credit} = 0$), the structural equation for check assigns $P(\text{check} = 1) = 0.2$: in 80% of noise realizations she is not checked at all, making $\text{loan} = 0$ certain without credit ever entering the causal chain. The PSCM makes this policy-determined bottleneck explicit and manipulable.

Definition 7 (Interventions) Given a model $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, an intervention on variables $\mathbf{X} = \{X_1, \dots, X_k\} \subseteq \mathbf{V}$ with values $\mathbf{x} = (x_1, \dots, x_k)$ replaces the structural assignments for X_i by $X_i := x_i$, where $i = 1, \dots, k$. We denote this intervention by either $[\mathbf{X} \leftarrow \mathbf{x}]$ or $\text{do}(\mathbf{X} = \mathbf{x})$.

Informally, an intervention surgically replaces a variable’s structural equation with a fixed constant, disconnecting it from its usual causes. The rest of the model propagates downstream as normal under the new assignment; the exogenous noise is unchanged.

In the OBCB model, $\text{do}(\text{gender} = 1)$ replaces Alice’s gender assignment with male, leaving all other structural equations intact. The downstream effect propagates: check now draws from $\text{Bern}(0.9)$ rather than $\text{Bern}(0.2)$, so Alice would be checked in 90% of noise realizations instead of 20%. This is the counterfactual world in which we ask whether her loan outcome would have differed.

3. Working within the PSCM framework commits the practitioner to fully specified structural assignments $\mathbf{F}$ and an exogenous distribution $P_{\mathbf{U}}$. This is a standard feature of probabilistic model-based causal inference — analogous to how Bayesian inference requires a likelihood and a prior — and it is precisely what enables PCI to yield individual-level counterfactual attributions rather than population-level bounds. Sampling from $P_{\mathbf{U}}$ is then a routine probabilistic inference operation. This requirement is a scope limitation worth stating explicitly. Standard Shapley values (Lundberg and Lee, 2017) require only expectations of the form $\mathbb{E}[f(\mathbf{X}_S, \mathbf{X}_{\bar{S}})]$ — no causal structure at all. Causal Shapley methods (Heskes et al., 2020) additionally require a causal graph and observational or interventional data, but not the structural equations themselves. Pearl’s PN/PS/PNS can be bounded from observational or interventional data given the graph, again without the full noise model. PCI requires strictly more: the complete structural assignments $\mathbf{F}$ and the noise distribution $P_{\mathbf{U}}$. The payoff is individual-level counterfactual attribution — not population-level bounds or marginal contributions averaged over a reference distribution — but practitioners without a fully specified structural model will need to build or assume one.

Because PCI supports arbitrary variable types, it will be helpful to define a measure theoretic object corresponding to interventional outcomes. Because outcomes are a deterministic function of exogenous noise, this will be a Dirac measure.

Definition 8 (Interventional Law) *Let $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ be a probabilistic structural causal model and let $Y \in \mathbf{V}$. For an intervention $\text{do}(\mathbf{X} = \mathbf{x})$, let $Y' : \text{dom}(\mathbf{U}) \rightarrow \text{dom}(Y)$ denote the measurable map induced by the modified structural assignments. The interventional law of Y given fixed $\mathbf{u} \in \text{dom}(\mathbf{U})$ is the probability measure on $\text{dom}(Y)$ defined by*

$$P(Y \in A \mid \mathbf{u}, \text{do}(\mathbf{X} = \mathbf{x})) := \mathbb{I}\{Y'(\mathbf{u}) \in A\} = \delta_{Y'(\mathbf{u})}(A), \quad A \subseteq \text{dom}(Y).$$

Informally, the interventional law records the deterministic outcome of Y under $\text{do}(\mathbf{X} = \mathbf{x})$ for a fixed noise realization $\mathbf{u}$: since the modified structural equations make Y a function of $\mathbf{u}$ alone, its distribution collapses to a point mass. Integrating over $P_{\mathbf{U}}$ then recovers the full interventional distribution.

We define a *causal set* to be a set of assignments to the variables of a probabilistic structural causal model.

Definition 9 (Causal set) *Given a model $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, a causal set is a set of pairs $X = x$ where $X \in \mathbf{V}$ and $x \in \text{dom}(X)$.*

Informally, a causal set is a list of specific variable-value assignments — a precise statement of which variables are set to which values. It is the basic unit of intervention: from a causal set one can read off exactly what is being manipulated or held fixed.

In what follows, causal sets will take on multiple roles. At times, they will represent sets of *potential* causes—whose subsets may be intervened upon—in which case we call them *suspects*, denoted $\mathbf{S}$. When referring to the specific subset we *in fact* intervene on in a given search step, we use $\mathbf{C} \subseteq \mathbf{S}$ and call them *active suspects*. Analogously, some variables serve as *potential* context variables to be held fixed at their actual values; we call these *witnesses*, denoted $\mathbf{W}$. The subset of witnesses actually held fixed in a given step is denoted $\mathbf{T} \subseteq \mathbf{W}$ and called *active witnesses*. When a variable appears in both $\mathbf{C}$ and $\mathbf{W}$, suspect activation takes precedence: $\mathbf{T}$ is pruned so that $\mathbf{T} \cap \mathbf{C} = \emptyset$.

To evaluate the causal impact of a variable on an outcome, we consider two interventional regimes: the *necessity regime*, in which we intervene on the variables $\mathbf{C} \subseteq \mathbf{S}$ to have an alternative value, and the *sufficiency regime*, in which we intervene so that $\mathbf{C}$ takes on factual values. Moreover, we hold some elements $\mathbf{T} \subseteq \mathbf{W}$ of the context fixed (i.e., intervened on to retain factual values, despite intervention on $\mathbf{C}$). When looking for an explanation, we perform a search across possible cause sets and possible contexts. In each case, attribution increases when the outcome is far from the factual value in the necessity world and decreases when it is far from the factual value in the sufficiency world.

Informally, the suspect set $\mathbf{S}$ names every variable that could plausibly have caused the outcome; the witness set $\mathbf{W}$ names context variables to be held fixed, isolating the causal channel under investigation. Intervening on suspects tests what would have happened under an alternative cause; fixing witnesses blocks confounding paths that would otherwise absorb the effect.

For both Alice (female, bad credit, loan denied) and Bob (male, bad credit, loan denied) we set $\mathbf{S} = \{\text{gender}, \text{credit}\}$ and $\mathbf{W} = \{\text{check-failed}\}$. The witness *check-failed* records whether the applicant was checked and had bad credit; holding it at its factual value isolates the causal path from *gender* or *credit* to *loan* independently of the credit-check bottleneck. Without this witness, intervening on *credit* for Alice would leave 80% of noise realizations unaffected — those in which she is never checked — making credit’s causal role invisible. The witness makes the bottleneck explicit and manipulable.

We expand an original model of interest with auxiliary sites that correspond to the moves involved in the search.

Definition 10 (Variable Selection Distribution) *Let $\mathbf{X}$ be a finite set of random variables. A variable selection distribution Γ is a probability distribution on $2^{\mathbf{X}}$. We write $\mathbf{Y} \sim \Gamma$ to mean $\mathbf{Y}$ is a random subset of $\mathbf{X}$ drawn from Γ , with probability mass function $p^\Gamma(\mathbf{y}) = \mathbb{P}(\mathbf{Y} = \mathbf{y})$ for $\mathbf{y} \subseteq \mathbf{X}$.*

Informally, the variable selection distribution Γ encodes prior beliefs about which subsets of suspects are worth examining. A uniform distribution treats all suspect subsets as equally plausible; a cardinality-constrained choice focuses the search on interactions of a particular order, analogous to choosing which interaction terms to include in a regression.

In particular, we will be using variable selection distributions for the powerset of causal suspects $\mathbf{S}$ and for the powerset of witnesses $\mathbf{W}$. One simple pattern of suspect and witness selection can be derived by setting size limits on the number of activated suspects or witnesses, and sampling uniformly otherwise.

Example 4 (Cardinality-Constrained Uniform Selection) *Let $\mathbf{X}$ be a candidate set of random variables and let $k \sim \text{Unif}\{J, \dots, K\}$ with $1 \leq J < K \leq |\mathbf{X}|$. A variable selection $\mathbf{Y} \sim \Gamma(2^{\mathbf{X}})$ is sampled from a cardinality-constrained uniform selection distribution $\Gamma(2^{\mathbf{X}})$ with probability mass function*

$$p^\Gamma(\mathbf{Y}) \propto \mathbb{I}[|\mathbf{Y}| = k].$$

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4. When $J = 1$ and $K = |\mathbf{X}|$, this distribution coincides with the Shapley weighting kernel, which samples coalition size uniformly and then selects a coalition uniformly at that size (Shapley, 1953). The parameters J and K allow the practitioner to restrict the search to cause sets of a specific cardinality range: setting $J = K = 1$ confines attribution to individual variables, while $J = 2, K = 3$ focuses on small multi-variable interactions. Moreover, whereas exact Shapley computation requires summing over all $2^{|\mathbf{X}|}$ coalitions, PCI treats Γ as a generative model: approximation proceeds by sampling subsets from Γ and outcomes from $P_{\mathbf{U}}$, making Monte Carlo estimation a native feature of the framework rather than an external algorithmic layer. We note that the weighting alone does not distinguish PCI from SHAP at $J = 1, K = |\mathbf{X}|$: the subset weights are identical. The distinction lies in the integrand, and it matters for what is being evaluated and what results follow. SHAP averages marginal contributions using the observational conditional $\mathbb{E}[f(\mathbf{X}_S = \mathbf{x}_S, \mathbf{X}_{\bar{S}})]$, marginalising over the empirical distribution of non-suspect features; this means a variable that is merely correlated with a cause — but not itself causally active — can receive positive attribution. PCI instead evaluates counterfactual outcomes $P(Y \in A \mid \mathbf{u}, \text{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{t}^*))$ computed via the structural model, so attribution flows only along genuine causal paths. The witness mechanism adds a further distinction: by holding intermediate variables fixed, PCI can resolve overdetermination and undercutting scenarios where SHAP assigns equal or incorrect attribution. The choice of J and K remains a design decision; practitioners can set $J = K = 1$ for single-variable attribution or allow higher-order interactions as discussed above. A detailed comparison of PCI, SHAP, and Causal SHAP on concrete examples is given in Section 4.

For the OBCB analysis we use a uniform Γ_s over the three non-empty subsets of $\mathbf{S}$: $\{gender\}$, $\{credit\}$, and $\{gender, credit\}$, each with probability $1/3$. With only two suspects there is no prior reason to privilege any combination, so the uniform choice is the unique symmetric default. Γ_w is uniform over $\{\emptyset, \{check-failed\}\}$, each with probability $1/2$. In larger suspect sets the choice of cardinality range becomes substantive: sampling uniformly from the full powerset yields an expected intervention size of $|\mathbf{S}|/2$, which spreads attribution across large coalitions and may make individual feature contributions harder to isolate; we return to this design decision in the experimental section.

Causal attribution hinges on questions such as “if X had, contrary to fact, been fixed to $x' \neq x$ instead of x , what would the outcome have been?” For non-binary causes, this is ambiguous. To resolve this in a way that does not require committing to a single, contrastive causal event (Kawakami et al., 2024), we rely on a *distribution* over alternative values.

We will assume the model provides us with a family of interventional distributions/densities (SCMs are an example of a model class satisfying this requirement), and explain how those are to be used to build up expanded distributions needed for causal impact estimation.

Definition 11 (Alternative Value Distribution) *Let $\mathbf{X}$ be a set of random variables with domain $\text{dom}(\mathbf{X})$ and let $\mathbf{X} = \mathbf{x} \in \text{dom}(\mathbf{X})$ be a factual event. An alternative value distribution $\Delta(\mathbf{x})$ is a probability measure on $\text{dom}(\mathbf{X})$ such that $\mathbf{x} \notin \text{supp}(\Delta(\mathbf{x}))$.*

Informally, $\Delta(\mathbf{x})$ specifies what counts as a genuinely different value for a suspect variable. The support condition ensures no mass falls on the factual value $\mathbf{x}$ itself, so every sampled alternative is a real counterfactual departure rather than a noisy copy of what actually happened.

One natural alternative value distribution results from taking the one currently embodied in the model and excising it around the factual setting. Then, the semantics for “alternative value” means “a sufficiently different value, following the original distribution otherwise”. In the discrete case, of course, the analogue is simply the probability mass function conditioned on the factual value not holding.

Example 5 (ϵ -Excised Distribution) *Assume $\mathbf{X}$ is $\mathbb{R}^d$ -valued with density p . Given a factual event $\mathbf{X} = \mathbf{x}$, define the ϵ -excised alternative value distribution $\Delta_\epsilon(\mathbf{x})$ as the probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ with density*

$$p_{\Delta_\epsilon}(\mathbf{x}' | \mathbf{x}) \propto \begin{cases} 0, & \mathbf{x}' \in B_\epsilon(\mathbf{x}), \\ p(\mathbf{x}'), & \text{otherwise,} \end{cases}$$

where $B_\epsilon(\mathbf{x})$ denotes the ϵ -ball centered at $\mathbf{x}$. In particular, $\mathbf{x} \notin \text{supp}(\Delta_\epsilon(\mathbf{x}))$.

Informally, the alternative value distribution Δ operationalises what “counterfactually different” means for each suspect variable: it specifies which alternative values are considered when asking whether the outcome would have changed under an alternative cause.

In the OBCB model both *gender* and *credit* are binary, so Δ is trivial. For Alice, the only alternative to *gender* = 0 (female) is *gender* = 1 (male), and the only alternative to *credit* = 0 (bad) is *credit* = 1 (good); for Bob, the only alternative to *gender* = 1 is *gender* = 0. Accordingly, $\Delta(\text{gender} = 0)$ is a point mass on *gender* = 1 and $\Delta(\text{credit} = 0)$ a point mass on *credit* = 1 — no distributional choice is required. The binary case is exactly where Δ is unambiguous; it

is for continuous suspects such as income or loan amount that the ϵ -excised distribution becomes essential, forcing an explicit commitment to what counts as a meaningfully different value.

5

We are now ready to state the main definitions of PCI. Because PCI is not constrained to particular variable types, we write the following in general measure theoretic terms that support combinations of continuous and discrete variables. Below, we use the interventional law (definition 8) — a Dirac measure on deterministic causal outcomes — and marginalize over random samples of suspect sets, witness sets, and alternative values for the causal set. This results in a joint measure on the outcome under the factual and counterfactual settings.

Definition 12 (Joint Necessity and Sufficiency Measure) *Fix a probabilistic structural causal model $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ and an outcome variable Y with domain $\text{dom}(Y)$. Let p_s^Γ and p_w^Γ be the mass functions of variable selection distributions on $2^{\mathbf{S}}$ and $2^{\mathbf{W}}$, and let $\Delta(\mathbf{s}^*)$ be an alternative value distribution on $\text{dom}(\mathbf{S})$. Let $(\mathbf{S}, \mathbf{W}) = (\mathbf{s}^*, \mathbf{w}^*)$ be a factual event.*

For a variable of interest X_k , let $2_k^{\mathbf{S}} := \{\mathbf{C} \subseteq \mathbf{S} : X_k \in \mathbf{C}\}$, and for each $\mathbf{C} \subseteq \mathbf{S}$ let $\Delta_{\mathbf{C}}(\mathbf{s}^)$ denote the pushforward of $\Delta(\mathbf{s}^*)$ under the restriction map $\mathbf{s}' \mapsto \mathbf{s}'|_{\mathbf{C}}$.⁶ For any measurable $A \subseteq \text{dom}(Y)$ and exogenous realization $\mathbf{u}$, the sufficiency and necessity measures are defined as follows. The sufficiency measure $P_k^s(A | \mathbf{u})$ intervenes on each active suspect set $\mathbf{C}$ to its factual values $\mathbf{s}^*|_{\mathbf{C}}$, holding active witnesses $\mathbf{T}$ fixed, and asks how likely $Y \in A$ is under that restoration. The necessity measure $P_k^n(A | \mathbf{u})$ instead integrates over alternative values $\mathbf{c}'$ for $\mathbf{C}$, again holding*

5. Any method that asks “what would happen if $\mathbf{X}$ had been different” must operationalise what *different* means. Using the empirical distribution $p(\mathbf{x}')$ directly — as in kernel SHAP (Lundberg and Lee, 2017) — is not a neutral choice: because p places mass near the factual value $\mathbf{x}$, sampled alternatives may be practically indistinguishable from $\mathbf{x}$, conflating the necessity question (“what if $\mathbf{x}$ had genuinely differed?”) with evaluating essentially the same setting. More generally, three natural choices exist for Δ . The observational marginal $p(\mathbf{x}')$ conflates the necessity question with the factual setting, as noted above. The fully conditional posterior $p(x'_k | \mathbf{x}_{-k})$ conditions on all other observed features — potentially including descendants of X_k — which imposes structural constraints that may rule out otherwise plausible counterfactuals, and still concentrates mass near x_k when features are correlated. The ϵ -excised distribution occupies the middle ground: it draws from the marginal informed by upstream variables only, excising the factual neighbourhood to guarantee genuine departure without downstream conditioning. The ϵ -excised distribution makes the commitment explicit and tunable: *different* means outside $B_\epsilon(\mathbf{x})$, with ϵ set by the practitioner to reflect the minimum change deemed meaningful. This parallels the Region of Practical Equivalence (ROPE) in Bayesian hypothesis testing (Kruschke, 2018): just as ROPE forces an explicit decision about what difference is large enough to matter, ϵ -excision forces an explicit decision about what counterfactual is sufficiently far from the factual to count as genuinely alternative. The practitioner makes this commitment regardless of method; the only question is whether it is made transparently or absorbed into a default. For practical guidance on choosing ϵ : when X is standardised to $\mathcal{N}(0, 1)$, the choice acquires a direct probabilistic interpretation — the excised mass $\Phi(x^* + \epsilon) - \Phi(x^* - \epsilon)$ is the fraction of the distribution ruled out as insufficiently different from the factual value x^* . At $x^* = 0$, setting $\epsilon = 0.2$ excises approximately 15.9% of the distribution, $\epsilon = 0.5$ excises 38.3%, and $\epsilon = 0.8$ excises 57.6%, corresponding roughly to small, medium, and large effect thresholds in the sense of Cohen (1988). Alternatively, ϵ can reflect measurement precision (the smallest meaningful increment of X) or a decision-relevant threshold (the minimum change in X that would alter a downstream outcome). As with ROPE, a sensitivity analysis across a range of ϵ values is advisable when the appropriate scale is unclear. For concrete examples of how SHAP’s implicit choice leads to problematic attributions, see Section 4.

6. For an assignment $\mathbf{x}$ to a collection of variables $\mathbf{X}$ and any subset $\mathbf{Y} \subseteq \mathbf{X}$, the restriction $\mathbf{x}|_{\mathbf{Y}}$ denotes the sub-assignment of $\mathbf{x}$ to the variables in $\mathbf{Y}$.

$\mathbf{T}$ fixed at factual values, and asks how likely $Y \in A$ would be had the causes genuinely differed.

$$P_k^s(A | \mathbf{u}) = \sum_{\mathbf{C} \in 2_k^{\mathbf{S}}} \sum_{\substack{\mathbf{T} \in 2^{\mathbf{W}} \\ \mathbf{T} \cap \mathbf{C} = \emptyset}} P(Y \in A | \mathbf{u}, \text{do}(\mathbf{C} = \mathbf{s}^* |_{\mathbf{C}}, \mathbf{T} = \mathbf{w}^* |_{\mathbf{T}})) p_s^\Gamma(\mathbf{C}) p_w^\Gamma(\mathbf{T}), \quad (1)$$

$$P_k^n(A | \mathbf{u}) = \sum_{\mathbf{C} \in 2_k^{\mathbf{S}}} \sum_{\substack{\mathbf{T} \in 2^{\mathbf{W}} \\ \mathbf{T} \cap \mathbf{C} = \emptyset}} \int P(Y \in A | \mathbf{u}, \text{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{w}^* |_{\mathbf{T}})) \Delta_{\mathbf{C}}(\mathbf{s}^*)(d\mathbf{c}') p_s^\Gamma(\mathbf{C}) p_w^\Gamma(\mathbf{T}). \quad (2)$$

The joint necessity-sufficiency measure then is given by

$$P_k^{s,n}(A \times B) := \int P_k^s(A | \mathbf{u}) P_k^n(B | \mathbf{u}) P_{\mathbf{U}}(d\mathbf{u}). \quad (3)$$

Informally, P_k^s asks: for a fixed noise realization $\mathbf{u}$, how likely is the outcome when causes take their factual values? P_k^n asks: how likely would the outcome be under alternative cause values? Their product, integrated over $P_{\mathbf{U}}$, captures both whether the actual cause was sufficient to produce the outcome and whether it was necessary — i.e. whether removing it would have changed things.

Applied to the OBCB model with the choices established above — $\mathbf{S} = \{\text{gender}, \text{credit}\}$, $\mathbf{W} = \{\text{check-failed}\}$, Γ_s and Γ_w uniform, Δ point masses on alternative binary values, outcome $y^* = \text{loan} = 0$ — Table 2 reports the expected causal impact for Alice and Bob under PNS, PCI without witnesses, and PCI with the *check-failed* witness active.

Person	Feature	PNS	PCI (no witnesses)	PCI (with witnesses)
Alice	<i>gender</i>	0.045	0.210	0.263
Alice	<i>credit</i>	0.180	0.240	0.199
Bob	<i>gender</i>	0.000	0.038	0.019
Bob	<i>credit</i>	0.860	0.228	0.119

Table 2: Expected causal impact of *gender* and *credit* on *loan* = 0 for Alice (female, bad credit) and Bob (male, bad credit) in the stochastic OBCB model. PNS uses no suspect or witness structure; PCI without witnesses sets $\mathbf{W} = \emptyset$; PCI with witnesses sets $\mathbf{W} = \{\text{check-failed}\}$.

For Alice, PNS and PCI without witnesses both fail **D-A-rank** (p. 9): they rank *credit* above *gender* (0.180 vs. 0.045; 0.240 vs. 0.210). PCI with witnesses reverses this (0.263 vs. 0.199), satisfying **D-A-rank** as well as **D-A1** and **D-A2** (both values positive). Holding *check-failed* at its factual value blocks the credit-check bottleneck in noise realizations where Alice was checked, isolating *gender*’s structural role. The *check-failed* witness is the decisive ingredient.

For Bob, all three methods satisfy **D-B1**, **D-B2**, and **D-B-rank**: *credit* is ranked above *gender* throughout. PNS assigns *gender* exactly 0.000 — satisfying **D-B1** only in the weak sense — while PCI with witnesses yields 0.019, giving *gender* a small positive score that reflects its role in opening the credit evaluation. The cross-individual desideratum **D-comp** holds under both PCI (0.263 > 0.019) and PNS (0.045 > 0.000). A full verification of all seven desiderata is given at the end of this section.

With the necessity-sufficiency measure fully defined, we are ready to characterize the causal impact of an event X_k . In particular, we will proceed by defining an arbitrary *causal impact function*, whose expectation over the necessity-sufficiency measure determines the causal impact. We first proceed with a formal definition of the causal impact measure, followed by several intuitive candidates for the causal impact function.

Definition 13 (Causal Impact Function and Its Expectation) *Let $y^* \in \text{dom}(Y)$ denote the factual outcome value, and let $Y^s, Y^n \sim P_k^{s,n}$. For any measurable function $ci : \text{dom}(Y)^3 \rightarrow \mathbb{R}$, the (expected) causal impact of X_k is defined as*

$$\mathbb{E}[ci(Y^s, Y^n, y^*)] := \iint ci(y^s, y^n, y^*) P_k^{s,n}(dy^s, dy^n). \quad (4)$$

Informally, ci is a user-defined scoring rule that translates the joint sufficiency–necessity measure into a single scalar. The choice is not arbitrary: the Absolute Difference example below is an instance of L1 distance between outcomes, the same metric underlying Average Treatment Effect, Conditional Treatment Effect, and SHAP. The binary indicator score recovers this as a special case when outcomes are binary — indicators are L1 on $\{0, 1\}$. Practitioners with reasons to prefer L2 or another distance measure may substitute it freely; the framework places no restriction on ci beyond measurability.

For the OCB analysis we instantiate ci as the PNS binary score defined in the following example, with $y^* = \text{loan} = 0$. This is the choice that produces the values in Table 2.

Example 6 (Probability of Necessary and Sufficient Causation for Binary Outcomes) *Assume the outcome Y is binary with $\text{dom}(Y) = \{0, 1\}$, and let $y^* = 1$ denote the factual outcome. Let Y^s and Y^n denote the sufficiency and necessity outcome variables. Define the causal impact score*

$$ci(y^s, y^n, y^*) = \mathbb{I}\{y^n \neq y^*\} \mathbb{I}\{y^s = y^*\}.$$

Then the expected causal impact reduces to

$$\mathbb{E}[ci(Y^s, Y^n, 1)] = \mathbb{P}(Y^n = 0, Y^s = 1),$$

which conceptually coincides with Pearl’s probability of necessity and sufficiency (PNS). The alignment is exact when $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\text{dom}(X_k) = \{0, 1\}$ where, factually, $x_k^ = 1$.*

The causal impact function ci is a user-defined component of PCI, and different choices encode different causal questions. In the benchmarking experiments of Section ??, we instantiate ci using only the necessity component, since Halpern’s actual causality has no sufficiency counterpart — necessity-only attribution allows a direct parallel comparison. The full framework incorporating both Y^s and Y^n is instantiated in the PNS binary example above and in the Absolute Difference example below; the OCB analysis of this section (Tables 2–3) provides a concrete validation, since the PNS binary ci jointly conditions on $Y^s = 1$ and $Y^n \neq y^*$ and it is precisely this combination that produces the correct **D-A-rank** ordering. Further experiments using the full ci are presented in Section ??.

Add cross-reference to sufficiency experiments once written.

Example 7 (Absolute Difference Impact Score) We can develop an analogous measure for continuous outcomes where we increase attribution with distance from the factual value y^* in the necessity world and decrease impact with distance to the factual value in the sufficiency world.

$$ci(y^s, y^n, y^*) = |y^n - y^*| - |y^s - y^*|.$$

With the full machinery in place — structural model, variable selection, alternative value distribution, joint necessity-sufficiency measure, and causal impact function — PCI delivers the following properties.

- (i) **Generalisation of PN/PS/PNS.** When $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\text{dom}(X_k) = \{0, 1\}$, the PNS binary ci recovers Pearl’s probability of necessity and sufficiency exactly; PN and PS are recovered by the corresponding one-sided ci choices.
- (ii) **Connection to actual causality.** Under suitable conditions on Γ and Δ , PCI recovers Halpern’s actual causality judgements; formal results are given in Section ??.
- (iii) **Necessity–sufficiency decomposition.** P_k^s and P_k^n are defined and interpretable independently; practitioners may weight them differently or use one alone depending on the causal question.
- (iv) **Witness-mediated symmetry breaking.** Holding intermediate variables fixed via $\mathbf{W}$ disambiguates overdetermination and undercutting scenarios where necessity-only methods assign equal or zero attribution, as demonstrated above for Alice and Bob.
- (v) **Composability.** Γ , Δ , and ci are probability distributions and a measurable function respectively; they slot naturally into Bayesian pipelines, accept posteriors as inputs, and admit Monte Carlo estimation without modification to the core formalism.

Table 3 verifies the seven desiderata introduced at the start of this section (p. 9) against the values in Table 2. PCI with the *check-failed* witness satisfies all seven; PNS fails only **D-A-rank**, the substantive ranking constraint for Alice.

Desideratum	PCI (with witnesses)	PNS
D-A1: $R(\text{gender} \mid \text{Alice}) > 0$	✓ 0.263	✓ 0.045
D-A2: $R(\text{credit} \mid \text{Alice}) > 0$	✓ 0.199	✓ 0.180
D-A-rank: gender > credit for Alice	✓ 0.263 > 0.199	× 0.045 < 0.180
D-B1: $R(\text{gender} \mid \text{Bob}) \geq 0$	✓ 0.019	✓ 0.000
D-B2: $R(\text{credit} \mid \text{Bob}) > 0$	✓ 0.119	✓ 0.860
D-B-rank: credit > gender for Bob	✓ 0.119 > 0.019	✓ 0.860 > 0.000
D-comp: Alice-gender > Bob-gender	✓ 0.263 > 0.019	✓ 0.045 > 0.000

Table 3: Verification of the seven desiderata (p. 9) against the values in Table 2. PCI with the *check-failed* witness satisfies all seven; PNS fails **D-A-rank**.

The single failure of PNS is **D-A-rank**: without a witness to hold the credit-check bottleneck fixed, necessity-based attribution cannot distinguish Alice’s gender (which blocked the evaluation entirely) from her credit (which would only have mattered had she been evaluated). The *check-failed* witness is what makes this distinction visible.

4. Examples and comparison to SHAP and Causal SHAP

Shapley values have become one of the most widely used tools for explaining the predictions of complex machine learning models. Grounded in cooperative game theory, they provide a principled decomposition of a model’s prediction into additive contributions from individual features. However, when the goal is *causal responsibility attribution* — explaining not just what a model predicts but why an outcome occurred — standard Shapley values suffer from fundamental limitations that are not always made explicit in the literature.

update section description when done

In this section we inspect these limitations carefully: we work throughout with a simple but instructive running example, simple enough to permit exact analytic computation while being rich enough to expose the core conceptual issues. We then examine how Causal Shapley values (Heskes et al., 2020) partially address these issues, where they succeed, and where fundamental gaps remain.

Example 8 (Signal with mediation) *We work with the following **generative model**, where X sends a signal, M mediates it with some noise, and Y is the final outcome, which is also noisy:*

$$X \sim \mathcal{N}(0.5, 0.25) \tag{5}$$

$$M = X + \varepsilon_M, \quad \varepsilon_M \sim \mathcal{N}(0, 0.1) \tag{6}$$

$$Y = M + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0, 0.1) \tag{7}$$

The causal graph is $X \rightarrow M \rightarrow Y$, a simple mediation chain. All noise terms are independent of each other and of X . Since everything is jointly Gaussian, all conditional expectations are linear functions of the conditioning variables — no approximations are required.

*Let $R(X \rightsquigarrow Y)$ denote the causal responsibility of X for Y . The **qualitative causal responsibility desiderata** for this example are rather simple:*

$$\mathbf{DXY} \quad R(X \rightsquigarrow Y) > 0$$

$$\mathbf{DMY} \quad R(M \rightsquigarrow Y) > 0$$

$$\mathbf{DXM} \quad R(X \rightsquigarrow M) > 0$$

$$\mathbf{DYX} \quad R(Y \rightsquigarrow X) = 0$$

$$\mathbf{DYM} \quad R(Y \rightsquigarrow M) = 0$$

$$\mathbf{DMX} \quad R(M \rightsquigarrow X) = 0$$

A somewhat more subtle desideratum is as follows:

$$\mathbf{DMXY} \quad R(M \rightsquigarrow Y) > R(X \rightsquigarrow Y)$$

this captures the intuition that responsibility should be higher for variables that are closer to the outcome in the causal graph.

*The key **distributional facts** that we will use are as follows:*

$$\mathbb{E}[X] = \mathbb{E}[M] = \mathbb{E}[Y] = 0.5, \tag{8}$$

$$(\text{Var}(X), \text{Var}(M), \text{Var}(Y)) = (0.25, 0.35, 0.45), \tag{9}$$

$$(\text{Cov}(X, M), \text{Cov}(X, Y), \text{Cov}(M, Y)) = (0.25, 0.25, 0.35). \tag{10}$$

For each target variable, the optimal predictor under squared loss is the conditional expectation, which can be computed exactly in this Gaussian model.

$$f_Y(X, M) = \mathbb{E}[Y \mid X, M] = M \quad (\{X, M\} \Rightarrow Y) \quad (11)$$

$$f_M(X, Y) = \mathbb{E}[M \mid X, Y] = 0.5X + 0.5Y \quad (\{X, Y\} \Rightarrow M) \quad (12)$$

$$f_X(M, Y) = \mathbb{E}[X \mid M, Y] = 0.5 + \frac{5}{7}(M - 0.5) \quad (\{M, Y\} \Rightarrow X) \quad (13)$$

add fn with derivations?

4.1. SHAP values for the signal with mediation example

Before we compute SHAP values for the case at hand, let's talk through the definition of **Shapley values**.

Definition 14 (SHAP) A cooperative game is a pair (N, v) , where $N = \{1, \dots, n\}$ is the set of players and $v : 2^N \rightarrow \mathbb{R}$ is the characteristic function mapping each subset (coalition) $S \subseteq N$ to the value that coalition can produce. The Shapley value (Shapley, 1953) is the unique attribution satisfying four axioms:

1. **Efficiency:** $\sum_i \phi_i = v(N) - v(\emptyset)$
2. **Symmetry:** If $v(S \cup \{i\}) = v(S \cup \{j\})$ for all S , then $\phi_i = \phi_j$
3. **Dummy:** If $v(S \cup \{i\}) = v(S)$ for all S , then $\phi_i = 0$
4. **Linearity:** For two games v_1, v_2 : $\phi_i(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 \phi_i(v_1) + \alpha_2 \phi_i(v_2)$

The unique solution is:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \quad (14)$$

The weight $\frac{|S|! (|N| - |S| - 1)!}{|N|!}$ equals the probability that, in a uniformly random ordering of all players, exactly the members of S precede i . Equivalently, ϕ_i is the average marginal contribution of i across all orderings. Lundberg and Lee (2017) apply Shapley values to explain model predictions by taking players to be input features, and the characteristic function to be the expected model output given a subset of features:

$$v(S) = \mathbb{E}_{X_{\bar{S}}} [f(X_S = x_S, X_{\bar{S}})] \quad (15)$$

where $\bar{S} = N \setminus S$ and missing features are marginalized.

Get back to this and illustrate:

One crucial observation here is that SHAP explains predictions, not outcomes. If $f(x) = \mathbb{E}[Y \mid X = x]$ (as in regression), then SHAP explains $\mathbb{E}[Y \mid X = x] - \mathbb{E}[Y]$. The residual $Y - f(x)$ — the noise — is entirely invisible to SHAP by construction.

Now consider a situation in which $X = 1, M = 1, Y = 1$. For each of those outcomes we will use our straightforward causal responsibility attribution desiderata and see how they align with SHAP values. First, let's compute the SHAP values for X and M with respect to the prediction of Y .

Target Y , Features $\{X, M\}$:

$$v(\emptyset) = \mathbb{E}[M] = 0.5 \quad (16)$$

$$v(\{X\}) = \mathbb{E}[M \mid X = 1] = 1 \quad (17)$$

$$v(\{M\}) = \mathbb{E}[M \mid M = 1] = 1 \quad (18)$$

$$v(\{X, M\}) = f_Y(1, 1) = 1 \quad (19)$$

Applying equation (14) with $|N| = 2$:

$$\phi_X = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 1) = 0.25 \quad \phi_M = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 1) = 0.25 \quad (20)$$

add which desiderata this violates

Target M , Features $\{X, Y\}$:

$$v(\emptyset) = 0.5 \quad (21)$$

$$v(\{X\}) = \mathbb{E}[0.5X + 0.5Y \mid X = 1] = 0.5 + 0.5 \cdot \mathbb{E}[Y \mid X = 1] = 1 \quad (22)$$

$$v(\{Y\}) = \mathbb{E}[0.5X + 0.5Y \mid Y = 1] \quad (23)$$

$$= 0.5 \cdot \mathbb{E}[X \mid Y = 1] + 0.5 = 0.5 \cdot 0.778 + 0.5 = 0.889$$

$$v(\{X, Y\}) = 1 \quad (24)$$

where $\mathbb{E}[X \mid Y = 1] = 0.5 + \frac{0.25}{0.45}(0.5) = 0.778$.

$$\phi_X = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 0.889) = 0.306 \quad \phi_Y = \frac{1}{2}(0.889 - 0.5) + \frac{1}{2}(1 - 1) = 0.194 \quad (25)$$

add which desiderata this violates

Target X , Features $\{M, Y\}$:

$$v(\emptyset) = 0.5 \quad (26)$$

$$v(\{M\}) = 0.5 + \frac{5}{7}(0.5) = 0.857 \quad (27)$$

$$v(\{Y\}) = \mathbb{E}[X \mid Y = 1] = 0.5 + \frac{0.25}{0.45}(0.5) = 0.778 \quad (28)$$

$$v(\{M, Y\}) = \mathbb{E}[X \mid M = 1, Y = 1] = 0.857 \quad (\text{since } X \perp Y \mid M) \quad (29)$$

$$\phi_M = \frac{1}{2}(0.857 - 0.5) + \frac{1}{2}(0.857 - 0.778) = 0.219 \quad (30)$$

$$\phi_Y = \frac{1}{2}(0.778 - 0.5) + \frac{1}{2}(0.857 - 0.857) = 0.139 \quad (31)$$

add which desiderata this violates

Summing up, the SHAP values for the three targets are as follows:

Target	Method	ϕ_X	ϕ_M	ϕ_Y
Y	Plain SHAP	0.25	0.25	—
M	Plain SHAP	0.306	—	0.194
X	Plain SHAP	—	0.219	0.139

Table 4: Plain SHAP attributions for instance $X = 1, M = 1, Y = 1$. For target X , plain SHAP assigns $\phi_Y = 0.139 > 0$, even though Y does not cause X . This arises because $v(\{Y\}) = 0.778 > 0.5$: observing $Y = 1$ is statistically informative about $X = 1$ through the chain $X \rightarrow M \rightarrow Y$. SHAP cannot distinguish “ Y is informative about X ” from “ Y causes X ” because the value function is grounded in observational conditionals, which are symmetric under reversal of causal direction.

4.2. Causal SHAP values for the signal with mediation example

Standard Shapley marginalizes over the distribution of features outside of a given coalition. From the causal perspective, if a feature outside the coalition is a descendant of a feature inside the coalition, its marginal distribution is wrong — instead we should use a distribution that reflects the intervention on its ancestor. This is the motivation behind [Heskes et al. \(2020\)](#), who propose replacing the observational conditional in the value function with an interventional one using Pearl’s do-calculus:

Definition 15 (Causal SHAP)

$$v_{\text{causal}}(S) = \mathbb{E}[f(X) \mid \text{do}(X_S = x_S)] = \int P(X_{\bar{S}} \mid \text{do}(X_S = x_S)) f(X_{\bar{S}}, x_S) dX_{\bar{S}} \quad (32)$$

By the rules of do-calculus, intervening on X_S only affects the distribution of descendants of X_S . For non-descendants, $P(X_j \mid \text{do}(X_S = x_S)) = P(X_j)$. Therefore equation (32) is equivalent to:

$$v_{\text{causal}}(S) = \mathbb{E}_{X_{\bar{S}}, \text{non-desc} \sim P(\cdot), X_{\bar{S}}, \text{desc} \sim P(\cdot \mid \text{do}(X_S = x_S))} [f(X_{\bar{S}}, x_S)] \quad (33)$$

The Shapley formula (14) is otherwise unchanged. The method retains all four Shapley axioms.

Now let’s compute Causal SHAP values for the same example and see how they align with our desiderata.

Target Y , features $\{X, M\}$:

$$v_{\text{causal}}(\emptyset) = v(\emptyset) = \mathbb{E}[M] = 0.5 \quad (34)$$

$$v_{\text{causal}}(\{X\}) = \mathbb{E}_{M \sim P(M \mid \text{do}(X=1))} [M] \quad (35)$$

$$= \mathbb{E}[M \mid \text{do}(X = 1)] = 1 \quad (36)$$

$$v_{\text{causal}}(\{M\}) = \sum_x P(X = x) \mathbb{E}[M \mid \text{do}(M = 1), X = x] = 1 \quad (37)$$

$$v_{\text{causal}}(\{X, M\}) = v(\{X, M\}) = 1 \quad (38)$$

this requires a gloss step by step

As all the v values are the same as in the plain SHAP case, we get the same results as with plain SHAP: $\phi_X = \phi_M = 0.25$.

Target M , features $\{X, Y\}$: Y is a descendant of X , so when $X \in S$:

$$v_{\text{causal}}(\{X\}) = \mathbb{E}_{Y \sim P(Y|do(X=1))}[0.5 \cdot 1 + 0.5Y] \quad (39)$$

$$= 0.5 + 0.5 \cdot 1 = 1 \quad (40)$$

When $Y \in S$ without X : X is not a descendant of Y , so X is marginalized over $P(X)$:

$$v_{\text{causal}}(\{Y\}) = \mathbb{E}_{X \sim P(X)}[0.5X + 0.5 \cdot 1] \quad (41)$$

$$= 0.5 \cdot 0.5 + 0.5 = 0.75 \quad (42)$$

This differs from plain SHAP ($v_{\text{causal}}(\{Y\}) = 0.889$). Result:

$$\phi_X = 0.375, \quad \phi_Y = 0.125 \quad (43)$$

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